Signals: Continuous and Discrete

A signal is a function carrying information: a voltage over time, a brightness over space, a daily temperature over days. Continuous-time signals x(t) are defined for every real t; discrete-time signals x[n] are defined only at integer n, usually because something sampled an underlying quantity. Two tiny signals do outsized work. The unit impulse (delta) is the idealized instant of energy at time zero: in discrete time, delta[n] equals 1 at n = 0 and 0 elsewhere. The unit step u[n] is 0 before time zero and 1 after, and it switches things on. Any discrete signal can be written as a sum of scaled, shifted impulses, which is exactly why the impulse is the right probe for studying systems.

System Properties

A system transforms an input signal into an output signal. A system is linear when scaling and adding inputs scales and adds outputs the same way: superposition holds. It is time-invariant when the system behaves the same today as tomorrow: shifting the input only shifts the output. It is causal when the output at any time depends only on present and past inputs, never future ones, which every real-time physical system must satisfy. It is BIBO stable when every bounded input produces a bounded output, so the system cannot blow up in response to a finite disturbance. These properties matter because the intersection of the first two, the LTI system, admits a complete and elegant theory that the rest of the course builds on.

LTI Systems and the Impulse Response

For a linear time-invariant system, one measurement characterizes everything: the impulse response h, the output produced when the input is a unit impulse. The reasoning is direct. Any input can be decomposed into scaled, shifted impulses. Time invariance says a shifted impulse produces a shifted copy of h. Linearity says a weighted sum of impulses produces the same weighted sum of responses. Therefore the output for any input is completely determined by h; there is nothing else to know about the system. Two useful facts follow immediately: an LTI system is causal exactly when h is zero for negative time, and it is BIBO stable exactly when h is absolutely summable (or absolutely integrable in continuous time).

Convolution

The recipe from the previous section has a name: convolution. In discrete time, y[n] = sum over k of x[k] h[n - k], written y = x * h. Each input sample launches its own scaled, shifted copy of the impulse response, and the output at time n is the sum of all the echoes arriving at that moment. Convolution is commutative (x * h = h * x), associative, and distributive over addition, so cascaded LTI systems can be collapsed into a single system whose impulse response is the convolution of the stages, in any order. A helpful special case: convolving a signal with a shifted impulse delta[n - d] simply delays the signal by d samples, nothing more.

Fourier Series

A periodic signal with period T can be written as a sum of harmonically related complex exponentials: sinusoids at the fundamental frequency 1/T and integer multiples of it. The Fourier series coefficients say how much of each harmonic the signal contains, and they are found by correlating the signal against each harmonic over one period. Smooth signals need few harmonics; signals with sharp edges need many, because edges are built from rapidly oscillating components. This is more than a curiosity: complex exponentials are eigenfunctions of LTI systems, meaning a sinusoid in yields a sinusoid out at the same frequency, only rescaled and phase-shifted. Decomposing into sinusoids therefore turns an LTI system into simple per-frequency multiplication.

The Fourier Transform

Aperiodic signals get the same treatment by letting the period grow without bound; the discrete set of harmonics merges into a continuum. The Fourier transform X(jw) = integral of x(t) e^(-jwt) dt measures how much of each frequency w the signal contains, and the inverse transform rebuilds x(t) from its spectrum. The pair reveals a signal's frequency content: where its energy lives, how fast it can change, how wide its bandwidth is. Key properties carry the load in practice: convolution in time becomes multiplication in frequency, a time shift becomes a phase ramp, and time-scaling a signal inversely scales its spectrum, which is the precise version of the slogan that short events are wideband.

Frequency Response and Filtering

Feeding e^(jwt) into an LTI system returns H(jw) e^(jwt), where H(jw), the frequency response, is the Fourier transform of the impulse response. For a sinusoidal input, the magnitude |H(jw)| multiplies the amplitude and the angle of H(jw) adds to the phase; the frequency itself never changes. This single fact is the basis of filtering. A low-pass filter has |H| near 1 at low frequencies and near 0 at high frequencies, keeping slow trends while discarding rapid wiggles; a high-pass filter does the opposite, and a band-pass filter keeps a chosen band. Designing a filter is choosing an achievable H(jw) that treats the frequencies you care about gently and the frequencies you don't harshly.

Sampling and Aliasing

Sampling reads a continuous signal every T_s seconds, producing $x[n] = x(nT_s)$ at sampling rate $f_s = 1/T_s$. The sampling theorem gives the exact condition for losing nothing: if $x(t)$ contains no frequencies above B, then sampling at f_s greater than 2B (the Nyquist rate) allows perfect reconstruction of $x(t)$ from its samples. Sample too slowly and aliasing occurs: frequencies above $f_s/2$ are folded down and masquerade as lower frequencies, indistinguishable from genuine ones once the samples are taken. This is why a wagon wheel appears to spin backwards on film. Practical systems therefore place an anti-aliasing low-pass filter before the sampler, removing content above $f_s/2$ so that what remains is sampled faithfully.

The Laplace Transform

The Laplace transform X(s) = integral of x(t) e^(-st) dt generalizes the Fourier transform by using the complex frequency s = sigma + jw, which handles growing and decaying signals that have no Fourier transform. Each transform comes with a region of convergence (ROC), the set of s where the integral exists; the same algebraic X(s) with different ROCs describes different signals. For systems described by linear differential equations, the transfer function H(s) is a ratio of polynomials, and its poles, the roots of the denominator, control behavior. A causal system is stable exactly when all poles lie strictly in the left half of the s-plane, because each pole contributes a mode e^(pt) that decays only when the pole's real part is negative.

The Z-Transform

Discrete-time systems get the same machinery through the z-transform, X(z) = sum of x[n] z^(-n). The unit circle in the z-plane plays the role the imaginary axis plays for Laplace: evaluating X(z) on the unit circle gives the discrete-time Fourier transform, when it exists. For a causal discrete LTI system with rational transfer function H(z), stability requires every pole to lie strictly inside the unit circle, since each pole p contributes a mode p^n that decays only when the magnitude of p is less than 1. Digital filter design is largely the art of placing poles and zeros in the z-plane: poles near the unit circle create resonant peaks, and zeros on the circle carve out notches at chosen frequencies.